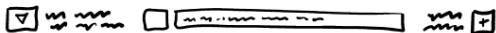


Special Relativity

PHYS505 – Electromagnetic Theory

Y. Torun

Illinois Institute of Technology



Black bean burrito bowl

Total time: 35 minutes

MY ACTUAL TIME: 1h 45m

$$t = \frac{t'}{\sqrt{1 - \frac{v^2}{c^2}}} \quad v = c \sqrt{1 - \left(\frac{t'}{t}\right)^2}$$

$$= c \sqrt{1 - \left(\frac{35}{105}\right)^2} = 0.94c$$

Tooltip: It says to cut the onions into 1/4" slices, but I'd better correct for length contraction.

I THINK THIS RECIPE AUTHOR IS MOVING
PAST ME AT 94% OF THE SPEED OF LIGHT.

Coordinate transformations – Galilean

- Galilean transformations between coordinates $(\vec{x}, t)$ in reference frame K and $(\vec{x}', t')$ in K' with constant relative velocity $\vec{v}$

$$\vec{x}' = \vec{x} - \vec{v}t, \quad t' = t$$

- Equations of motion for a system of particles with positions, velocities $\{\vec{x}_i\}, \{\vec{v}_i\}$ interacting through central two-body potential in frame K'

$$m_i \frac{d\vec{v}_i'}{dt'} = -\vec{\nabla}_i' \sum_{j \neq i} V_{ij}(|\vec{x}_i' - \vec{x}_j'|)$$

- Terms in the equation transform as

$$\frac{d}{dt'} = \frac{d}{dt}, \quad \vec{v}_i' = \vec{v}_i - \vec{v}, \quad \vec{\nabla}_i' = \vec{\nabla}_i, \quad |\vec{x}_i' - \vec{x}_j'| = |\vec{x}_i - \vec{x}_j|$$

- so it looks the same in frame K

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- For a field $\psi(\vec{x}', t')$ satisfying the wave equation in frame K'

$$\left(\nabla'^2 - \frac{1}{c^2} \partial_t'^2 \right) \psi'(\vec{x}', t') = 0$$

- The terms transform as

$$\nabla' = \nabla, \quad \partial_t' = \partial_t - \vec{v} \cdot \vec{\nabla}$$

- resulting in

$$\left\{ \nabla^2 - \frac{1}{c^2} \left[\partial_t^2 + 2 (\vec{v} \cdot \vec{\nabla}) \partial_t + (\vec{v} \cdot \vec{\nabla})^2 \right] \right\} \psi = 0$$

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Coordinate transformations – Lorentz

- For K' moving with $\vec{v}$ in K , we define

$$x_0 = ct, \quad \vec{\beta} = \frac{\vec{v}}{c}, \quad \beta = |\vec{\beta}|, \quad \gamma = \frac{1}{\sqrt{1 - \beta^2}}$$

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- and the Lorentz transformations read

$$x'_0 = \gamma (x_0 - \vec{\beta} \cdot \vec{x})$$

$$\vec{x}' = \vec{x} - \gamma \vec{\beta} x_0 + (\gamma - 1) (\hat{\beta} \cdot \vec{x}) \hat{\beta}$$

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- The components $\perp$ to $\vec{v}$ stay invariant

$$\vec{x}_{\parallel} = (\hat{\beta} \cdot \vec{x}) \hat{\beta}$$

$$\vec{x}_{\perp} = \vec{x} - \vec{x}_{\parallel}$$

$$\vec{x}'_{\parallel} = \gamma (\vec{x}_{\parallel} - \vec{\beta} x_0)$$

$$\vec{x}'_{\perp} = \vec{x}_{\perp}$$

$$x'_0 = \gamma (x_0 - \beta x_{\parallel})$$

$$\vec{x}' = \gamma (\vec{x}_{\parallel} - \vec{\beta} x_0) + \vec{x}_{\perp}$$

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- also expressed using the *boost parameter* or *rapidity*

$$\beta = \tanh \zeta, \quad \gamma = \cosh \zeta, \quad \beta\gamma = \sinh \zeta$$

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- which gives

$$\begin{aligned} x'_0 &= x_0 \cosh \zeta - x_{\parallel} \sinh \zeta \\ x'_{\parallel} &= -x_0 \sinh \zeta + x_{\parallel} \cosh \zeta \end{aligned} \qquad \vec{x}'_{\perp} = \vec{x}_{\perp}$$

4-vectors

- Defined as objects that transform like $(x_0, \vec{x})$

$$A'_0 = \gamma (A_0 - \vec{\beta} \cdot \vec{A})$$

$$A'_{\parallel} = \gamma (A_{\parallel} - \beta A_0)$$

$$\vec{A}'_{\perp} = \vec{A}_{\perp}$$

- The scalar product of 4-vectors, defined as

$$A_0 B_0 - \vec{A} \cdot \vec{B}$$

is invariant under Lorentz transformations

$$A'_0 B'_0 - \vec{A}' \cdot \vec{B}' = A_0 B_0 - \vec{A} \cdot \vec{B}$$

$$A_0^2 - |\vec{A}|^2 = A_0'^2 - |\vec{A}'|^2$$

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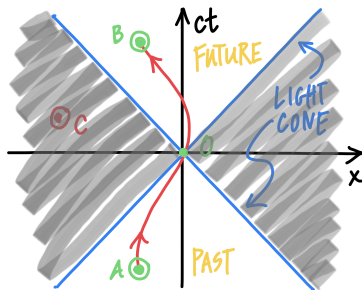
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Light cone

- Divides space-time into 3 regions with respect to the origin (here and now) based on the invariant interval $s^2 = (ct)^2 - |\vec{x}|^2$

- $s^2 > 0$ (time-like)
 - Past: $t < 0$ (OA)
 - Future: $t > 0$ (OB)
- $s^2 < 0$ (space-like)
 - Elsewhere (OC)

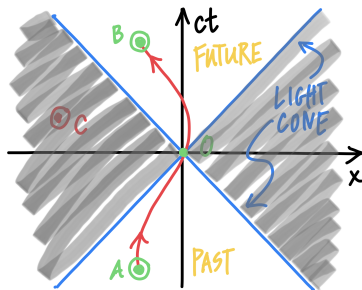


- Each point corresponds to an event
- Path of light represented by *light cone*
- World-line* (trajectory of system, eg. particle) stays within light cone
- Pairs of points with space-like separation (inside vs outside cone) are causally disconnected

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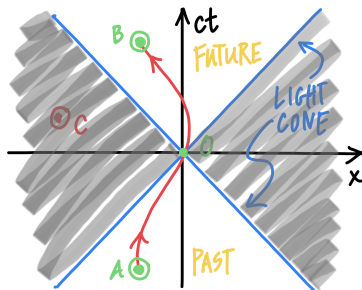


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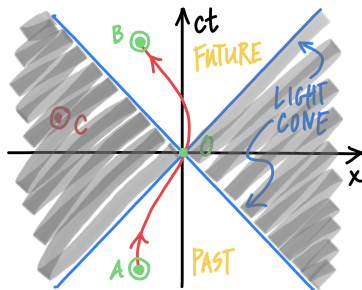


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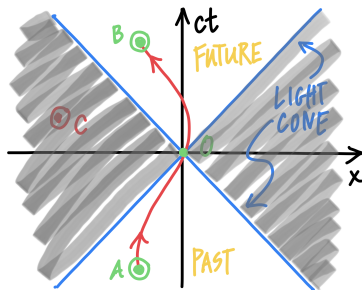


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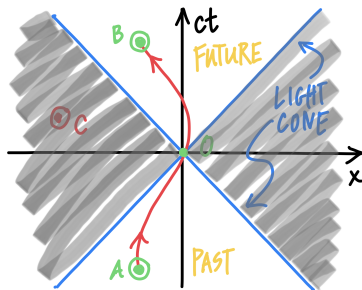


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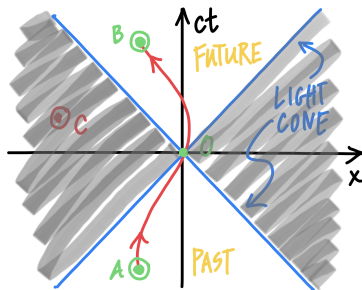


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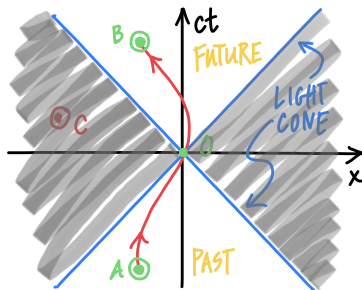


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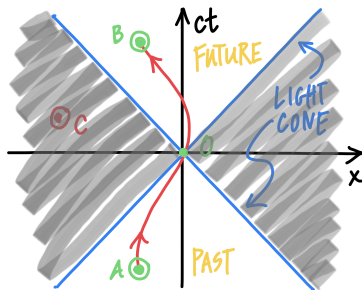


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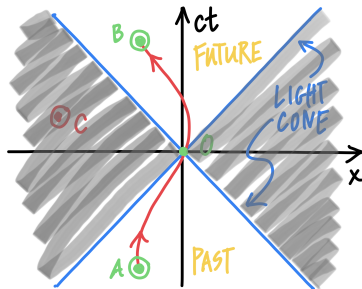
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Invariant interval

For two events $P_1(t_1, \vec{x}_1)$ and $P_2(t_2, \vec{x}_2)$ in reference frame K

$$s_{12}^2 = c^2(t_1 - t_2)^2 - |\vec{x}_1 - \vec{x}_2|^2$$

- If the separation is time-like ($s_{12}^2 > 0$), there exists a frame K' in which the two events happen at the same point at different times

$$\beta = \frac{|\vec{x}_1 - \vec{x}_2|}{c|t_1 - t_2|} < 1 \rightarrow \vec{x}'_1 = \vec{x}'_2, \quad [c(t'_1 - t'_2)]^2 = s_{12}^2$$

- If the separation is space-like ($s^2 < 0$), there exists a frame K'' in which the events happen at the same time at different locations

$$t''_1 = t''_2, \quad |\vec{x}''_1 - \vec{x}''_2|^2 = -s_{12}^2$$

- If the separation is light-like ($s_{12}^2 = 0$), the events are only connected by light signals

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Proper time

- For a system (eg. particle) in frame K with velocity $\vec{u}(t)$, the change in invariant interval

$$ds^2 = (c dt)^2 - |d\vec{x}|^2 = (c dt)^2 - |\vec{u} dt|^2 = (c dt)^2(1 - \beta^2)$$

with $\beta = u/c$

- Proper time* interval $d\tau$ is Lorentz invariant

$$d\tau = dt \sqrt{1 - \beta(t)^2} = \frac{dt}{\gamma(t)}, \quad ds = c d\tau$$

- and corresponds to the time interval in the rest frame of the system
- with *time dilatation* in frame K

$$t_2 - t_1 = \int_{\tau_1}^{\tau_2} \frac{d\tau}{\sqrt{1 - \beta^2(\tau)}} = \int_{\tau_1}^{\tau_2} \gamma(\tau) d\tau$$

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$$t_2 - t_1 = \int_{\tau_1}^{\tau_2} \frac{d\tau}{\sqrt{1 - \beta^2(\tau)}} = \int_{\tau_1}^{\tau_2} \gamma(\tau) d\tau$$

Doppler shift

- A plane wave of frequency ω and wavevector $\vec{k}$ in frame K will have the same phase in frame K'

$$\phi = \omega t - \vec{k} \cdot \vec{x} = \omega' t' - \vec{k}' \cdot \vec{x}'$$

- $\omega = ck_0$ and $\vec{k}$ transform as a 4-vector

$$k'_0 = \gamma (k_0 - \vec{\beta} \cdot \vec{k}), \quad k'_{\parallel} = \gamma (k_{\parallel} - \beta k_0), \quad \vec{k}'_{\perp} = \vec{k}_{\perp}$$

- For light waves

$$\omega' = \gamma \omega (1 - \beta \cos \theta), \quad \tan \theta' = \frac{\sin \theta}{\gamma (\cos \theta - \beta)}$$

where $\cos \theta = \hat{\mathbf{k}} \cdot \hat{\mathbf{v}}$

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Addition of velocities

- For a point moving with velocity $\vec{u}'$ in frame K' which is moving with velocity $\vec{v}$ with respect to frame K , the components of the velocity $\vec{u}$ in frame K are

$$u_{\parallel} = \frac{u'_{\parallel} + v}{1 + \frac{\vec{v} \cdot \vec{u}'}{c^2}}, \quad \vec{u}_{\perp} = \frac{\vec{u}'_{\perp}}{\gamma_v \left(1 + \frac{\vec{v} \cdot \vec{u}'}{c^2}\right)}$$

- which reduces to the Galilean result $\vec{u} = \vec{u}' + \vec{v}$ for $u', v \ll c$
- For parallel $\vec{u}', \vec{v}$

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- Ordinary velocities ($d\vec{x}/dt$) don't transform as 4-vectors

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- but the combination $(\gamma_u c, \gamma_u \vec{u})$ does

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Energy, momentum

- Relativistic momentum for a particle of mass m and velocity $\vec{u}$

$$\vec{p} = \gamma m \vec{u} = \frac{m \vec{u}}{\sqrt{1 - \frac{u^2}{c^2}}}$$

- and its total energy

$$E = \gamma mc^2 = \frac{mc^2}{\sqrt{1 - \frac{u^2}{c^2}}}$$

- form a 4-vector $(p_0 = E/c, \vec{p})$ with invariant length $(mc)^2$

$$p_0^2 - \vec{p} \cdot \vec{p} = (mc)^2, \quad E = \sqrt{(pc)^2 + (mc^2)^2}$$

- We often specify numerical values with factors of c left out:
 $m_e \simeq 0.51 \text{ MeV}/c^2$, $p = 0.86 \text{ MeV}/c$, $E = 1.0 \text{ MeV}$

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$$\vec{p} = p_{\parallel} \hat{\mathbf{z}} + \vec{p}_{\perp}$$

in frame K

- there is a unique transformation to a frame K' where the axial momentum is 0

$$\vec{p}' = \vec{p}_{\perp}, \quad \frac{E'}{c} = \sqrt{p_{\perp}^2 + (mc^2)^2} = \Omega$$

- and the components in K can be expressed as

$$p_{\parallel} = \Omega \sinh \zeta, \quad \frac{E}{c} = \Omega \cosh \zeta$$

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Lorentz group

- Lorentz transformations for space-time coordinates $(x_0, \vec{x})$ preserve

$$s^2 = x_0^2 - \vec{x} \cdot \vec{x} = x_0^2 - x_1^2 - x_2^2 - x_3^2$$

- Together with rotations, the set of such transformations form the *homogeneous Lorentz group*
- Combined with translations and reflections in space and time, they form the *inhomogeneous Lorentz group* or *Poincaré* and preserve the interval between points x, y

$$s^2(x, y) = (x_0 - y_0)^2 - (x_1 - y_1)^2 - (x_2 - y_2)^2 - (x_3 - y_3)^2$$

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- For a transformation of 4-coordinates from x to x'

$$x'^{\alpha} = \Lambda^{\alpha}_{\beta} x^{\beta}, \quad \alpha = 0, 1, 2, 3$$

- For a transformation of 4-coordinates from x to x'

$$x'^{\alpha} = x'^{\alpha}(x^0, x^1, x^2, x^3), \quad \alpha = 0, 1, 2, 3$$

- Tensors defined in terms of their transformation properties, indicated by convention with the position of indices

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$$A'^{\alpha} = \frac{\partial x'^{\alpha}}{\partial x^{\beta}} A^{\beta}$$

with an implied sum over repeated indices (Einstein convention)

Covariance

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- For a covariant rank-2 tensor $G_{\alpha\beta}$

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- Higher rank and mixed indices also defined, eg.

$$H'^{\alpha}{}_{\beta\gamma} = \frac{\partial x'^{\alpha}}{\partial x^{\lambda}} \frac{\partial x^{\mu}}{\partial x'^{\beta}} \frac{\partial x^{\nu}}{\partial x'^{\gamma}} H^{\lambda}{}_{\mu\nu}$$

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- Scalar product defined as

$$A \cdot B = A_{\alpha} B^{\alpha}$$

is invariant

- *Metric tensor* g defines geometry of space-time

$$(ds)^2 = g_{\alpha\beta} dx^\alpha dx^\beta$$

Metric of space-time

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- and connects covariant/contravariant quantities

$$g^{\alpha\beta} = g_{\alpha\beta}$$

$$x^\alpha = g^{\alpha\beta} x_\beta$$

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- Partial derivatives wrt contra/co-variant coordinates transform as co/contra-variant vectors

$$\frac{\partial}{\partial x'^\alpha} = \frac{\partial x^\beta}{\partial x'^\alpha} \frac{\partial}{\partial x^\beta}$$

Metric of space-time

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$$\begin{aligned}g^{\alpha\beta} &= g_{\alpha\beta} & g_{\alpha\gamma} g^{\gamma\beta} &= \delta_{\alpha}^{\beta} \\x^{\alpha} &= g^{\alpha\beta} x_{\beta} & x_{\alpha} &= g_{\alpha\beta} x^{\beta} \\A^{\alpha} &= (A^0, \vec{A}) & A_{\alpha} &= (A^0, -\vec{A})\end{aligned}$$

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- and a 4-D derivative operator can be defined as

$$\partial_{\alpha} \equiv \frac{\partial}{\partial x^{\alpha}} = \left(\frac{\partial}{\partial x_0}, \vec{\nabla} \right), \quad \partial^{\alpha} \equiv \frac{\partial}{\partial x_{\alpha}} = \left(\frac{\partial}{\partial x^0}, -\vec{\nabla} \right)$$

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- with a corresponding 4-divergence and 4-Laplacian (aka d'Alembertian)

$$\partial_\alpha A^\alpha = \partial^\alpha A_\alpha = \frac{\partial A^0}{\partial x^0} + \vec{\nabla} \cdot \vec{A}$$
$$\square \equiv \partial_\alpha \partial^\alpha = \frac{\partial^2}{\partial x^{02}} - \nabla^2$$

Matrix representation of Lorentz group

- Using a column vector for the components of a 4-vector

$$x = \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}, \quad g = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad gx = \begin{pmatrix} x^0 \\ -x^1 \\ -x^2 \\ -x^3 \end{pmatrix}$$

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$$a \cdot b = \tilde{a} g b = (a^0 a^1 a^2 a^3) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} b^0 \\ b^1 \\ b^2 \\ b^3 \end{pmatrix}$$

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$$a \cdot b = \tilde{a} g b = (a^0 a^1 a^2 a^3) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} b^0 \\ b^1 \\ b^2 \\ b^3 \end{pmatrix}$$

- Lorentz transformations are represented by matrices A that preserve the 4-D norm

$$x' = Ax, \quad x' \cdot x' = x \cdot x \rightarrow \tilde{A} g A = g \rightarrow \det A = \pm 1$$

Matrix representation of Lorentz group

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- where L is real and traceless and gL is antisymmetric

$$g\tilde{A}g = A^{-1}, \quad g^2 = I, \quad \det g = -1, \quad \tilde{g}L = -gL$$

$$L = \begin{pmatrix} 0 & L_{01} & L_{02} & L_{03} \\ L_{01} & 0 & L_{12} & L_{13} \\ L_{02} & -L_{12} & 0 & L_{23} \\ L_{03} & -L_{13} & -L_{23} & 0 \end{pmatrix}$$

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$$S_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad e^{-\theta S_1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \theta & \sin \theta \\ 0 & 0 & -\sin \theta & \cos \theta \end{pmatrix}$$

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$$S_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad e^{-\theta S_2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & 0 & -\sin \theta \\ 0 & 0 & 1 & 0 \\ 0 & \sin \theta & 0 & \cos \theta \end{pmatrix}$$

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$$L = -\vec{\omega} \cdot \vec{S} - \vec{\zeta} \cdot \vec{K}, \quad A = e^L$$

with $\vec{\omega}, \vec{\zeta}$ containing the 6 parameters

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- The matrices S, K are a representation of the *infinitesimal generators* of the Lorentz group and satisfy

$$[S_i, S_j] = \epsilon_{ijk} S_k, \quad [S_i, K_j] = \epsilon_{ijk} K_k, \quad [K_i, K_j] = -\epsilon_{ijk} S_k$$

SI $\leftarrow$ Units in electromagnetism $\rightarrow$ Gaussian

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$$\epsilon_0 = \frac{10^7}{4\pi c^2}$$

$$\mu_0 = 4\pi \times 10^{-7}$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}$$

$$\vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \partial_t \vec{D}$$

$$\vec{\nabla} \times \vec{E} = -\partial_t \vec{B}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{F} = q\left(\vec{E} + \frac{1}{c}\vec{v} \times \vec{B}\right)$$

$$\epsilon_0 = 1$$

$$\mu_0 = 1$$

$$\vec{D} = \vec{E} + 4\pi\vec{P}$$

$$\vec{H} = \vec{B} - 4\pi\vec{M}$$

$$\vec{\nabla} \cdot \vec{D} = 4\pi\rho$$

$$\vec{\nabla} \times \vec{H} = \frac{4\pi}{c}\vec{J} + \frac{1}{c}\partial_t \vec{D}$$

$$\vec{\nabla} \times \vec{E} = -\frac{1}{c}\partial_t \vec{B}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

SI vs Gaussian units

- Continuity eqn looks the same

$$\partial_t \rho + \vec{\nabla} \cdot \vec{J} = 0$$

- No fudge factors for vacuum in Gaussian units, 4π from geometry

$$E = \frac{q}{r^2}, \quad \vec{\nabla} \cdot \vec{E} = 4\pi\rho$$

- Fields ($\vec{E}, \vec{D}, \vec{B}, \vec{H}$) all have the same dimension
- Slap a factor of 4π in front of "sources" ($\rho, \vec{J}, \vec{P}, \vec{M}$) and use factors of c to make up any difference in dimensions

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Covariant formulation of electrodynamics

- Charge density $\rho(\vec{x}, t)$ and current density $\vec{J}(\vec{x}, t)$ form a 4-vector

$$J^\alpha = (c\rho, \vec{J})$$
$$\partial_\alpha J^\alpha = \partial_t \rho + \vec{\nabla} \cdot \vec{J} = 0$$

- The scalar and vector potentials also form a 4-vector

$$A^\alpha = (\Phi, \vec{A})$$
$$\partial_\alpha A^\alpha = \frac{1}{c} \partial_t \Phi + \vec{\nabla} \cdot \vec{A} = 0$$
$$\square A^\alpha = \frac{4\pi}{c} A^\alpha$$

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Covariant formulation of electrodynamics

- The fields form the components of an antisymmetric 2nd rank field-strength tensor

$$F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha, \quad F^{\alpha\beta} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix}$$

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- The covariant components are obtained by $\vec{E} \rightarrow -\vec{E}$

$$F_{\alpha\beta} = g_{\alpha\gamma} F^{\gamma\delta} g_{\delta\beta} = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & -B_y & B_x & 0 \end{pmatrix}$$

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- The dual field-strength tensor is defined as

$$\mathcal{F}^{\alpha\beta} = \frac{1}{2} \epsilon^{\alpha\beta\gamma\delta} F_{\gamma\delta}$$

and can be obtained from $F^{\alpha\beta}$ by $\vec{E} \rightarrow \vec{B}, \vec{B} \rightarrow -\vec{E}$

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- Inhomogeneous eqns for $\vec{E}, \vec{B}$ can be written in terms of F

$$\vec{\nabla} \cdot \vec{E} = 4\pi\rho, \quad \vec{\nabla} \times \vec{B} - \frac{1}{c}\partial_t\vec{E} = \frac{4\pi}{c}\vec{J} \rightarrow \partial_\alpha F^{\alpha\beta} = \frac{4\pi}{c}J^\beta$$

- while the homogeneous eqns become

$$\vec{\nabla} \cdot \vec{B} = 0, \quad \vec{\nabla} \times \vec{E} + \frac{1}{c}\partial_t\vec{B} = 0 \rightarrow \partial_\alpha \mathcal{F}^{\alpha\beta} = 0$$

- or, in terms of F

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0$$

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$$\partial_\alpha G^{\alpha\beta} = \frac{4\pi}{c} J^\beta, \quad \partial_\alpha \mathcal{F}^{\alpha\beta} = 0$$

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$$\frac{dp^\alpha}{d\tau} = m \frac{dU^\alpha}{d\tau} = \frac{q}{c} F^{\alpha\beta} U_\beta$$

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- A pure $\vec{E}$ - or $\vec{B}$ -field in K turns into a mixture in K'

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$$F'^{\alpha\beta} = \frac{\partial x'^{\alpha}}{\partial x^{\gamma}} \frac{\partial x'^{\beta}}{\partial x^{\delta}} F^{\gamma\delta}$$

- and are, in vector form

$$\vec{E}' = \gamma \left(\vec{E} + \vec{\beta} \times \vec{B} \right) - \frac{\gamma^2}{\gamma + 1} \left(\vec{\beta} \cdot \vec{E} \right) \vec{\beta}$$

$$\vec{B}' = \gamma \left(\vec{B} - \vec{\beta} \times \vec{E} \right) - \frac{\gamma^2}{\gamma + 1} \left(\vec{\beta} \cdot \vec{B} \right) \vec{\beta}$$

- A pure $\vec{E}$ - or $\vec{B}$ -field in K turns into a mixture in K'
- Only components $\perp \vec{\beta}$ are transformed

$$\hat{\beta} \cdot \vec{E}' = \hat{\beta} \cdot \vec{E}$$

$$\hat{\beta} \cdot \vec{B}' = \hat{\beta} \cdot \vec{B}$$

$$\vec{E}'_{\perp} = \gamma \left(\vec{E}_{\perp} + \vec{\beta} \times \vec{B}_{\perp} \right) \quad \vec{B}'_{\perp} = \gamma \left(\vec{B}_{\perp} - \vec{\beta} \times \vec{E}_{\perp} \right)$$

Transformation of fields

- and are, in vector form

$$\vec{E}' = \gamma \left(\vec{E} + \vec{\beta} \times \vec{B} \right) - \frac{\gamma^2}{\gamma + 1} \left(\vec{\beta} \cdot \vec{E} \right) \vec{\beta}$$

$$\vec{B}' = \gamma \left(\vec{B} - \vec{\beta} \times \vec{E} \right) - \frac{\gamma^2}{\gamma + 1} \left(\vec{\beta} \cdot \vec{B} \right) \vec{\beta}$$

- A pure $\vec{E}$ - or $\vec{B}$ -field in K turns into a mixture in K'
- Only components $\perp \vec{\beta}$ are transformed

$$\hat{\beta} \cdot \vec{E}' = \hat{\beta} \cdot \vec{E}$$

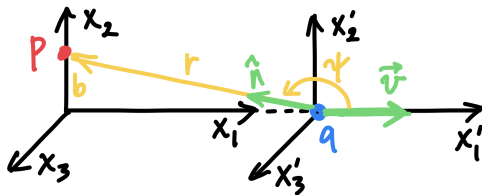
$$\hat{\beta} \cdot \vec{B}' = \hat{\beta} \cdot \vec{B}$$

$$\vec{E}'_{\perp} = \gamma \left(\vec{E}_{\perp} + \vec{\beta} \times \vec{B}_{\perp} \right) \quad \vec{B}'_{\perp} = \gamma \left(\vec{B}_{\perp} - \vec{\beta} \times \vec{E}_{\perp} \right)$$

- If charges at rest in K'

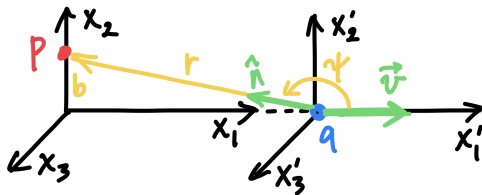
$$\vec{B}' = \vec{0} \rightarrow \vec{B} = \vec{\beta} \times \vec{E}$$

Field from moving charge



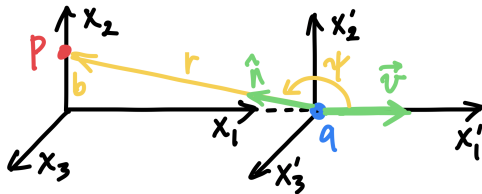
- q at $(0, 0, 0)$ in K'
- passes the origin in K at $t = t' = 0$
- Point $\vec{P}$ coordinates:
 $(0, b, 0)$ in K ,
 $(-\gamma vt, b, 0)$ in K'

Field from moving charge



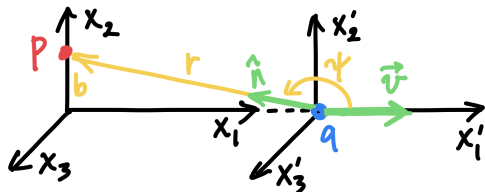
- q at $(0, 0, 0)$ in K'
- passes the origin in K at $t = t' = 0$
- Point $\vec{P}$ coordinates:
 $(0, b, 0)$ in K ,
 $(-\gamma vt, b, 0)$ in K'

Field from moving charge



- q at $(0, 0, 0)$ in K'
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 $(0, b, 0)$ in K ,
 $(-\gamma vt, b, 0)$ in K'

Field from moving charge

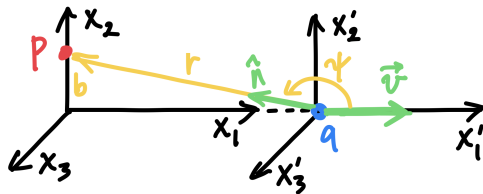


- q at $(0, 0, 0)$ in K'
- passes the origin in K at $t = t' = 0$
- Point $\vec{P}$ coordinates:
 $(0, b, 0)$ in K ,
 $(-\gamma vt, b, 0)$ in K'

- Field in rest frame K'

$$\vec{E}' = \frac{q}{r'^3} \vec{r}' = \frac{q}{[b^2 + (\gamma vt)^2]^{3/2}} [-\gamma vt \hat{x}_1 + b \hat{x}_2], \quad \vec{B}' = \vec{0}$$

Field from moving charge



- q at $(0, 0, 0)$ in K'
- passes the origin in K at $t = t' = 0$
- Point $\vec{P}$ coordinates:
 $(0, b, 0)$ in K ,
 $(-\gamma vt, b, 0)$ in K'

- Field in rest frame K'

$$\vec{E}' = \frac{q}{r'^3} \vec{r}' = \frac{q}{[b^2 + (\gamma vt)^2]^{3/2}} [-\gamma vt \hat{x}_1 + b \hat{x}_2], \quad \vec{B}' = \vec{0}$$

- Field at point P in K

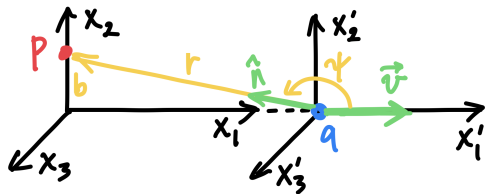
$$\vec{E} = \frac{q}{r^3 \gamma^2 (1 - \beta^2 \sin^2 \psi)^{3/2}} \vec{r} \quad \frac{E_1}{E_2} = -\frac{\gamma vt}{b}$$

$$E_1 = E'_1 = -\frac{q\gamma vt}{[b^2 + (\gamma vt)^2]^{3/2}} \quad E_2 = \gamma E'_2 = \frac{\gamma qb}{[b^2 + (\gamma vt)^2]^{3/2}}$$

$$E_3 = B_1 = B_2 = 0$$

$$B_3 = \gamma \beta E'_2 = \beta E_2$$

Field from moving charge



- q at $(0, 0, 0)$ in K'
- passes the origin in K at $t = t' = 0$
- Point $\vec{P}$ coordinates:
 $(0, b, 0)$ in K ,
 $(-\gamma vt, b, 0)$ in K'

- Field in rest frame K'

$$\vec{E}' = \frac{q}{r'^3} \vec{r}' = \frac{q}{[b^2 + (\gamma vt)^2]^{3/2}} [-\gamma vt \hat{x}_1 + b \hat{x}_2], \quad \vec{B}' = \vec{0}$$

- Field at point P in K

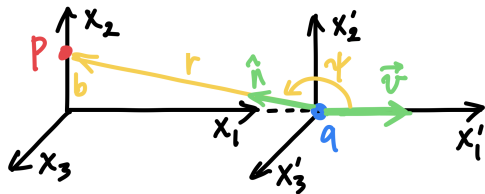
$$\vec{E} = \frac{q}{r^3 \gamma^2 (1 - \beta^2 \sin^2 \psi)^{3/2}} \vec{r} \quad \frac{E_1}{E_2} = -\frac{\gamma vt}{b}$$

$$E_1 = E'_1 = -\frac{q \gamma vt}{[b^2 + (\gamma vt)^2]^{3/2}} \quad E_2 = \gamma E'_2 = \frac{\gamma q b}{[b^2 + (\gamma vt)^2]^{3/2}}$$

$$E_3 = B_1 = B_2 = 0$$

$$B_3 = \gamma \beta E'_2 = \beta E_2$$

Field from moving charge



- q at $(0, 0, 0)$ in K'
- passes the origin in K at $t = t' = 0$
- Point $\vec{P}$ coordinates:
 $(0, b, 0)$ in K ,
 $(-\gamma vt, b, 0)$ in K'

- Field at point P in K , $\beta \rightarrow 0$

$$\vec{E} \rightarrow \frac{q}{r^3} \vec{r}$$

$$E_1 = E'_1 \rightarrow 0$$

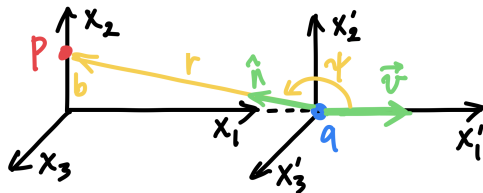
$$E_3 = B_1 = B_2 = 0$$

$$\frac{E_1}{E_2} = -\frac{\gamma vt}{b}$$

$$E_2 = \gamma E'_2 \rightarrow \frac{q}{b^2}$$

$$B_3 = \gamma \beta E'_2 \simeq \frac{q}{c} \frac{\vec{v} \times \vec{r}}{r^3}$$

Field from moving charge



- Field at point P in K , $\beta \rightarrow c$

$$\vec{E} \rightarrow \frac{q}{r^3 \gamma^2 (1 - \sin^2 \psi)^{3/2}} \vec{r}$$

$$E_1 = E'_1 \rightarrow 0$$

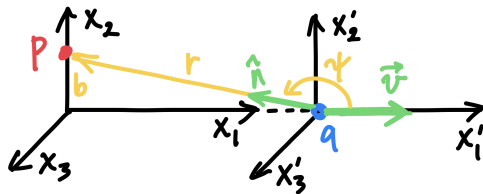
$$E_3 = B_1 = B_2 = 0$$

$$\frac{E_1}{E_2} = -\frac{\gamma vt}{b}$$

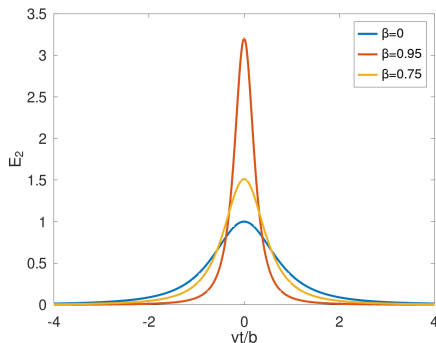
$$E_2 = \gamma E'_2 \rightarrow 0$$

$$B_3 = \gamma \beta E'_2 \rightarrow E_2$$

Field from moving charge



- q at $(0, 0, 0)$ in K'
- passes the origin in K at $t = t' = 0$
- Point $\vec{P}$ coordinates:
 $(0, b, 0)$ in K ,
 $(-\gamma vt, b, 0)$ in K'

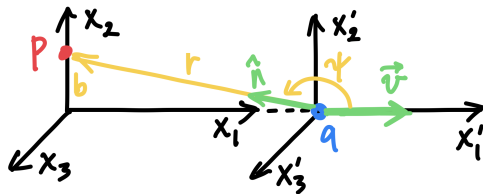


$$E_{\perp} = E_2 = \frac{1}{\beta} B_3$$

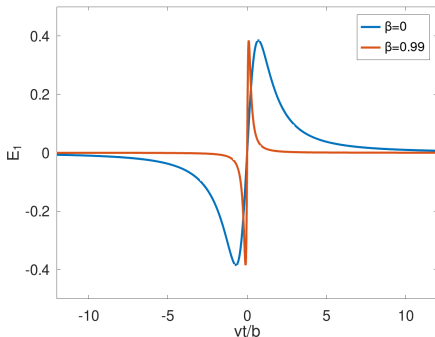
$$\text{peak } E_2(t = 0) = \gamma \frac{q}{b^2}$$

$$\text{width } \Delta t \approx \frac{b}{\gamma v}$$

Field from moving charge



- q at $(0, 0, 0)$ in K'
- passes the origin in K at $t = t' = 0$
- Point $\vec{P}$ coordinates:
 $(0, b, 0)$ in K ,
 $(-\gamma vt, b, 0)$ in K'



$$E_{\parallel} = E_1$$

$$\text{peak } |E_1(t = \pm \frac{1}{\sqrt{2}\gamma} \frac{b}{v})| = \frac{2}{3\sqrt{3}} \frac{q}{b^2}$$

$$\text{width } \Delta t \approx \frac{b}{\gamma v}$$